24.9.2026

Practical - 7

Sliding window

AIM:

Write program to implement flow control at data link layer using SLIDING WINDOW PROTOCOL. Simulate the flow of frames from one node to another.

Code:

sender.py

```
import time
window_size = int(input("Enter window size"))
message = input("Enter message")
frames = []
for i in range(0, len(message), 5):
    data = message[i:i+5]
    frames.append([i//5, data])
print("\nFrames created")
for frame in frames:
    print("Frame No:", frame[0], "Data:", frame[1])

start = 0
while start < len(frames):
    end = min(start + window_size, len(frames))
    print("\nsending frames...")
    with open("sender_Buffer.txt", "w") as file:
        for i in range(start, end):
            file.write(f"{frames[i][0]} {frames[i][1]}\n")
            print("sending Frame", frames[i][0], ":", frames[i][1]
    print("Waiting for ACK")
    time.sleep(2)
    try:
        with open("receiver_Buffer.txt", "r") as file:
            ack = int(file.read().strip())
        print("Ack received:", ack)
    except:
        print("NO ACK received")
        continue
```

```
if ack == end - 1:
    print("ACK is correct.")
    start = end
else:
    print("NACK / Wrong ACK")
    print("Resending frames from frame", ack+1)
    if ack >= start:
        start = ack + 1
    else:
        start = start
print("\n All frames transmitted successfully!")
```

Receiver.py

```
import time
import os
print("Receiver started ...")
expected_frame = 0
old_buffer = ""
while true:
    if not os.path.exists("sender_Buffer.txt"):
        time.sleep(1)
        continue
    with open("sender_Buffer.txt", "r") as file:
        data = file.read()
    if data == old_buffer:
        time.sleep(1)
        continue
    old_buffer = data
    lines = data.strip().split("\n")
    print("\n Received frames:")
    last_ack = expected_frame - 1
    for line in lines:
        parts = line.split(" ", 1)
        frame_no = int(parts[0])
```

```
            frame_data = parts[1]
            print("Frame", frame_no, ":", frame_data)
            if frame_no == expected_frame:
                print("Frame accepted")
                last_ack = frame_no
                expected_frame += 1
            else:
                print("Wrong Frame!")
                break
        print("Sending ACK:", last_ack)
        with open("receiver_buffer.txt", "w") as file:
            file.write(str(last_ack))
        last_frame = int(lines[-1].split(" ", 1)[0])
        if last_ack == last_frame:
            print("\nAll frames received successfully!")
            break
        time.sleep(1)
```

OUTPUT:

```
>> python sender.py
Enter window size: 3
Enter the message: Computer Networker
Frames created:
Frame No: 0 Data: compu
Frame No: 1 Data: ter N
Frame No: 2 Data: etwor
Frame No: 3 Data: ks

Sending Frames...
Sending Frame 0: compu
Sending Frame 1: ter N
Sending Frame 2: etwor

Waiting for ACK...
ACK received: 3
NACK / Wrong ACK!
```

Resending frames from Frame 4

All frames transmitted successfully!

>> python Receiver.py

Receiver started...

Received frames:

Frame 0 : compu

Frame accepted

Frame 1 : for N

Frame accepted

Frame 2 : etwork

Frame accepted

Sending ACK : 2

All frames received successfully!

RESULT:

The sliding window protocol was successfully implemented using Python. Frames were transmitted and acknowledged successfully, demonstrating reliable data transmission.